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MILITARY SPECIFICATION
IMAGE INTENSIFIER ASSEMBLY
25 MILLIMETER, MICROCHANNEL INVERTER
MX-9644/UV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the 25 Millimeter, Microchannel Inverter, Image Intensifier Assembly, MX-9644/UV.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2.e).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-ED-TM Fort Monmouth, NJ 07703-5023 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-P-11268	- Parts, Materials, and Processes Used in Electronic Equipment
MIL-B-49030/6	- Battery, Dry BA-3058/U
MIL-P-49090	- Power Supply, 25 Millimeter, Microchannel Inverter
MIL-B-49430/4	- Battery, Primary, Lithium, Sulfur-Dioxide BA-5567/U

STANDARDS

MILITARY

MIL-STD-202	- Test Methods for Electronics and Electrical component Parts
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-781	- Reliability Testing for Engineering Development, Qualification, and Production

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building #4, Section D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2.e).

DRAWINGS

USACECOM

SM-D-850311	- Tube Assembly, Image Intensifier, 25 MM Microchannel Inverter
SM-D-850310	- Tube Assembly Image Intensifier

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this specification takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Description. The Image Intensifier Assembly, 25 Millimeter, Microchannel Inverter MX-9644/UV, hereinafter designated as the assembly, shall consist of a single module containing a microchannel electron multiplier as the primary gain mechanism. The assembly shall employ an S20 photocathode with an extended red (ER) response, an aluminized phosphor screen and a power supply. The assembly shall be encapsulated and be provided with a hard surface insulating sleeve, or boot, and electrical contacts.

3.2 Construction. The assembly shall be in accordance with SM-D-850311 and all provisions of this specification.

3.2.1 Weight. The assembly weight shall not exceed 350 grams.

3.3 Qualification. Unless otherwise specified by the contract or purchase order, Image Intensifier Assemblies furnished under this specification shall be products approved as qualified products (see 3.6.27.16 and 6.3).

3.3.1 Initial Production Testing. When specified in the contract or purchase order (see 6.2.b), the contractor shall furnish ten (10) IPT units in accordance with 4.3.

3.4 Parts and materials. Parts and materials shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification and shall conform to MIL-P-11268. Non-standard parts lists are required for the power supply assembly, soldering material, wire, and non-corrosive potting materials.

3.5 Components.

3.5.1 Components. Components shall be as specified herein and as shown on the applicable drawings.

3.5.2 Power supply. The power supply shall be in accordance with MIL-P-49090, and as specified herein.

3.5.3 Manual gain control. The assembly shall operate in conjunction with a nominal 50 thousand ohm gain control switch, and a 470 thousand ohm resistor in accordance with Figure 1. The minimum manual gain setting shall be obtained with the switch (S1) closed, and the wiper rotated CCW to position 1. The maximum manual gain setting shall be obtained with the switch (S1) closed and the wiper rotated CW to position 3.

3.5.4 Voltage source. The voltage source for operating the assembly shall be from an BA-3058/U battery conforming to MIL-B-49030/6, BA-5567/U battery conforming to MIL-B-49430/4 or a power source with an input of 2.0 to 3.0 volts direct current (dc).

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3.6 Operation and environmental.

3.6.1 Photocathode sensitivity. The photocathode sensitivity shall be not less than the following values for the conditions specified.

a. Luminous sensitivity of 260 microamperes (uA) per lumen for 2856 kelvin (K) radiation.
b. Radiant sensitivity of 0.015 ampere per watt at 0.80 micrometer.
c. Radiant sensitivity of 0.010 ampere per watt at 0.85 micrometer.

3.6.2 Burn-in, environmental stress screening (ESS). The assembly shall undergo cyclic operation (burn-in) of not less than 120 continuous cycles. Each cycle of operation shall have a profile as specified in Figure 2. At the end of this cyclic operation there shall not be evidence of damage (see 3.6.27.1).

3.6.2.1 High temperture stress test. The assembly shall undergo a high temperature stress test to evaluate the photocathode stability after processing and the following post test parameters shall be met.

3.6.2.1.1 Photocathode sensitivity shall not decay more than 5% from the pretest readings and will not decay below the requirements of 3.6.1.a.

3.6.2.1.2 Photoresponse shall meet the requirements of 3.6.1.b and 3.6.1.c.

3.6.2.1.3 There shall be no evidence of damage (see 3.6.27.1).

3.6.2.2 High input illumination stress test. With operating potentials applied to the tube module illuminate the photocathode equivalent to 1×10^{-3} ft.c. for 48 hours. Photocathode sensitivity shall not decay more than 5% from pretest value and the tube shall meet the requirements of 3.6.1.

3.6.3 Vibration. With the operating potential applied the assembly shall not be damaged, (see 3.6.27.1) after being rigidly mounted, singly or in groups, and subjected to a 2.5 g's (see 3.6.27.3) peak sine wave vibration over a frequency range of 10 hertz (Hz) through 3500 Hz along both the longitudinal and transverse axes of the assembly.

3.6.4 Mechanical shock.

3.6.4.1 Mechanical shock I. With the operating potential applied, the assembly shall not be damaged (see 3.6.27.1) when subjected to 6 shock impacts applied parallel to the optical axes and 6 shock impacts applied perpendicular to the optical axis of the assembly. The shock impacts shall be nominal half sine wave pulse having a minimum of 500 g's (see 3.27.3) peak value and a duration of 0.3 milliseconds, ± 0.05 milliseconds measured between the 10 percent values of the peak amplitude.

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3.6.4.2 Mechanical shock II. The assembly shall not be damaged (see 3.6.27.1) when subjected to 6 mechanical shock pulses along the perpendicular axis and 6 mechanical shock pulses alone the parallel axis at an amplitude of 140 g's, ± 10 g's (see 3.6.27.3) half-sine wave peak for 9 milliseconds, ± 10 percent duration.

3.6.5 Environmental temperature. The assembly shall not be damaged (see 3.6.27.1) by the storage, operation, or thermal shock temperature profile specified in Figure 3 and shall meet the requirements at the temperature specified.

a. Temperature: plus 52°C

(1) Input current - Shall not exceed 35 milliamperes.

(2) Equivalent background input (EBI) - Not greater than 6 x 10^{-10} lumens/cm^2.

(3) Gain @ 1 to 5 x 10^{-6} footcandle (fc) - Not less than 20 thousand and not greater than 80 thousand.

(4) Screen brightness @ 5 x 10^{-4} to 1 to 10^{-3} fc - Not less than 2 footlamberts and not greater than 10 footlamberts.

(5) Screen brightness @ .01 to .03 fc - Not less than 2 footlamberts and not greater than 10 footlamberts.

(6) Operational fluctuation @ .05 to .1 fc shall be not greater than plus or minus 10 percent of steady state value for not less than 30 seconds.

(7) Rise time - Not greater than 5 seconds, overshoot less than 40 percent of steady state output brightness.

b. minus 54°C

(1) Input current - Not greater than 25 milliamperes.

(2) EBI - Not greater than 2 x 10^{-11} lumens/cm^2.

(3) Gain @ 1 to 5 x 10^{-6} fc - Not less than 20 thousand and not greater than 80 thousand.

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(4) Screen brightness @ 5 x 10^{-4} to 1 x 10^{-3} fc - Not less than 2 footlamberts and not greater than 10 footlamberts.

(5) Screen brightness @ .01 to .03 fc - Not less than 2 footlamberts and not greater than 10 footlamberts.

(6) Operational fluctuation @ .05 to .1 fc shall be not greater than plus or minus 10 percent of steady state value for not less than 30 seconds.

(7) Rise time - not greater than 5 seconds, overshoot less than 40 percent of steady state output brightness.

(8) The variation between room temperature gain tests prior to high temperature and after cold temperature shall not be greater than plus or minus 15 percent. The room temperature EBI shall be less than or equal to 2 x 10^{-11} lumens/cm^2 after high and low temperature test.

c. minus 17.8°C

(1) Gain @ 1 to 5 x 10^{-6} fc - Not less than 30 thousand and not greater than 70 thousand.

(2) Rise time - Not greater than 5 seconds.

3.6.6 Equivalent background input. The equivalent background input at room temperature shall not exceed 2 x 10^{-11} lumens/cm^2.

3.6.7 Luminance gain. The assembly shall have a room temperature (see 3.6.27.12) luminance gain and high light saturation characteristics as specified in Table I and Figures 4 and 5. The input current shall not exceed 32 milliamperes (mA) at the light levels specified.

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TABLE I. Saturation requirements for luminance gain.

Input Manual Gain Setting	Nominal Input Light Level (fc)	Minimum Allowable Gain	Maximum Allowable Gain	Minimum Allowable Output Brightness (ft-l)	Maximum Allowable Output Brightness (ft-l)	Maximum Input Current (mA)	Applicable Figure
Maximum	5×10^{-6}	40,000	70,000	--	--	32	4
Maximum	5 x 10-4	--	--	2	10	32	4
Maximum	5×10^{-2}	--	--	2	10	NA	4
Maximum	10.0	0	10^{-3}	--	--	NA	4
Maximum	5×10^{-5}	2,000	10,000	--	--	32	5
Maximum	5×10^{-3}	--	--	2	10	32	5

3.6.8 Time dependent bright source. When the assembly is subjected to a periodic pulsed input illumination in such a manner as to give rise to photocurrents in accordance with profiles (a) and (b) as shown in Figures 6 and 7, the assembly shall have response at maximum manual gain setting as set forth below.

3.6 8.1 Recovery time, profile (a) Figure 6. The assembly shall recover to within ± 50 percent of the ambient useful steady state output brightness (see 3.6.28.14) in 400 milliseconds or less after the end of the applied illumination, profile (a). Recovery time to the ambient useful steady state output brightness shall be not greater than 800 milliseconds.

3.6.8.2 Recovery time, profile (b), Figure 7. The assembly shall recover to within ± 50 percent of the ambient useful steady state brightness in 300 milliseconds or less after the end of the applied illumination profile (b). Recovery time to the ambient useful steady state output brightness shall be not greater than 600 milliseconds.

3.6 8.3 Resolution, profile (b) Figure 7. With a steady state illumination input of 1 to 5 x 10^{-5} footcandle and after 3 cycles of profile (b), the assembly shall have a minimum resolution of 16 line pairs per millimeter, (lp/mm) during continuous application of profile (b).

3.6.9 Bright source protection. The assembly shall not be damaged (see 3.6.27.1) when the input illumination is 50 millilumens concentrated on a 1.0 square millimeter area of the photocathode during a time interval of 20 seconds, ± 1 second.

3.6.10 Signal-to-noise ratio. The signal-to-noise ratio (S/N) of the assembly shall be not less than 2.8 to 1.

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3.6.11 Fixed pattern noise. Fixed pattern noise shall be as follows:

3.6 11.1 Multi-to-multi-pattern variation. Deviation from the mean output brightness shall not exceed plus or minus 10 percent.

3.6.11.2 Multi-boundary pattern noise. When sampled by a 25 micrometer photometering aperture the deviation from the mean output brightness over three boundaries shall be within the limits of plus 8 or minus 15 percent.

3.6.12 Output brightness uniformity. The output brightness uniformity of the assembly shall not exceed a ratio of 2 to 1 over the full 25 millimeter useful cathode diameter.

3.6.13 Image alignment. The center of an image produced on the screen of the assembly by focusing a test reticle on the photocathode concentric with the optical axis shall fall within a 0.040 inch diameter circle on the image screen concentric with the optical axis.

3.6.14 Image shift. The center of an image produced on the screen of the assembly by focusing a test reticle on the photocathode concentric with the optical axis shall shift not more than 0.005 inch from the optical axis.

3.6.15 Resolution.

3.6.15.1 Center resolution. The center resolution of the intensifier shall be not less than 32.0 lp/mm with an input radiation of 1×10^{-3} to 1×10^{-4} fc.

3.6.15.2 High light level center resolution. With the entire photocathode of the assembly illuminated with not less than 0.8 nor greater than 5 footcandles the minimum high light level center resolution shall be 16 lp/mm.

3.6.15.3 Peripheral resolution. The peripheral resolution of the assembly shall be not less than 28 lp/mm at four points, 10mm, ± 0.5mm from the center of the photocathode with an input radiation of 1×10^{-3} to 1×10^{-4} fc.

3.6.16 Modulation transfer function. With an input light level of less than or equal to 2×10^{-3} footcandle, the modulation transfer function (MTF) (see 3.6.27.6) at the center of the image screen shall be not less than :

a. 90 percent MTF at 2.5 lp/mm.
b. 60 percent MTF at 7.5 lp/mm.
c. 25 percent MTF at 15 lp/mm.

3.6.17 Useful cathode diameter. The useful cathode diameter shall be not less than 25 millimeters.

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3.6.18 Magnification. The magnification shall be not less than 0.96, nor greater than 1.04 over the useful cathode area. The magnification shall not vary more than 5 percent over the useful cathode area.

3.6.19 Shear distortion. Shear distortion (see 3.6.27.9) of the assembly shall not be greater than 50 micrometers at any point within 0.8 inch diameter quality circle concentric with optical axis of the assembly.

3.6.20 Electrical stability. The assembly shall show no evidence of damage when the operating assembly is probed with a 1/4 inch diameter ground probe. The input current shall not deviate more than 0.5 milliamperes.

3.6.21 Phosphor decay. With an input illumination of 1 to 5 x 10^{-5} footcandle, the decay due to the output phosphor screen response shall be within the limits specified in Figure 8 at 10 milliseconds, 100 milliseconds, 500 milliseconds, and 1 second from the start of decay.

3.6.22 Phosphor spectral response. The peak relative spectral response of the output screen shall occur at a wavelength between 5100 and 5600 angstroms. The bandwidth shall be less than 2000 angstroms measured at the 10 percent points of the spectral response curve. The relative response at 6500 angstroms shall be not greater than 10 percent of the nominal peak response.

3.6.23 Photocathode and screen quality. When the screen of the assembly is viewed with a 10X or greater magnifier and with the radiation level on the photocathode adjusted to obtain best spot contrast, the bright spots, opaque spots, or dark spots which exceed a contrast of 30 percent of their surrounding area shall not exceed the size and quantities specified in Table II. The size of non-circular spots shall be determined on the basis of equal area to circular spots. When the distance between two spots is less than the maximum dimension of either spot, the two spots shall be considered as one spot with size equal to the sum of the maximum dimensions of the two spots plus the amount of separation between them. Graininess shall not be discernible anywhere over the useful diameter of the screen. There shall be no field emissions (see 3.6.27.5) except as allowed in Table II or ion spots (see 3.6.27.4) discernible on the output phosphor screen when viewed with a 10X or greater magnifier after the input photocathode of the assembly has been illuminated with an initial level of not less than 5 x 10^{-3} footcandles and then reduced to 2 x 10^{-7} footcandles.

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TABLE II. Assembly spots.

Size of Spots In Inches	Accumulative Number of Spots Within 0.30 Inch Diameter Circle	Accumulative Number of Spots Within Area Bounded By Two Circles 0.30 and 0.80 Inches in Diameter	Accumulative Number of Spots Within Area Bounded By Two Circles 0.80 and 0.98 Inches in Diameter
Greater than 0.015 inch	0	0	0
0.012 - 0.015 inch	0	1	2
0.009 - 0.0119 inch	0	3	6
0.006 - 0.0089 inch	1	6	10
0.003 - 0.0059 inch	1	15	20
*0 - 0.0009 inch	1	3	5
*0.001 - 0.003 inch	1	3	4
*Greater than 0.003 inch	0	0	0

* Bright spots only

1. The 0.30, 0.080, 0.98 inch circles on the image screen shall be concentric with the optical axis of the assembly.

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3.6.24 Reliability. The assembly shall have a minimum mean-time-to failure (MTTF) of 2500 hours when operated under the reliability test conditions specified herein.

3.6.25 Identification markings. Each assembly shall be identified in accordance with drawing SM-D-850310. The markings shall include a coded acceptance date. The first two numbers of the code shall be two digits indicating the calendar week of the year (01 through 52). Reading from left to right, or, top to bottom, the code number shall indicate the year and week of acceptance in that order.

3.6.26 Workmanship. Workmanship shall conform to Requirement 9, MIL-STD-454.

3.6.27 Technical interpretations. The following technical interpretations are, when referenced in sections 3, 4, or 5, mandatory for this specification.

3.6.27.1 Damage. Damage is defined as electrical failure or malfunctioning including arcing, corona, flashing, bright spots more than allowed in 3.6.23, flickering, blinking, field emissions, ion spots, cracks, scratches, breakage, deformation, corrosion, or deterioration of any part or finish, (more than allowed in paragraph 4.6.4 and 4.6.4.1) or missing or loose components.

3.6.27.2 Optical length. The optical length of the assembly is defined as the distance from the input image plane to the output image plane.

3.6.27.3 "g". "g" is a unitless measure of the quantity of units of force associated with the earth gravity that a object is subjected to.

3.6.27.4 Ion spot. A high concentration of ion scintillations generally in the center of the field.

3.6.27.5 Discernible field emission. Discernible field emission is extraneous emission which appears as bright spots or pattern that may flicker or appear intermittently on the image screen in one general position. Field emission is voltage dependent and is best observed with no radiation incident on the photocathode.

3.6.27.6 Modulation transfer. Modulation transfer is a measure of the degradation of a signal image as it appears at the output screen of the assembly as correlated to the input signal pattern which is normalized to 100 percent contrast at a spatial frequency equal to or less than 0.2 line pair per millimeter.

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3.6.27.7 Limiting resolution. Limiting resolution is defined as the smallest resolution pattern which the observer can see and distinguish between the black lines and the clear area between the black lines. The observer must be able to determine the number of line pairs in both the vertical and horizontal test pattern.

3.6.27.8 Ion scintillations. Bright spots which occur on the image screen randomly in space and time.

3.6.27.9 Shear distortion. Shear distortion is defined as the displacement or rotation of a small line segment with respect to a perfect straight line image.

3.6.27.10 Output brightness. All output light levels refer to those levels at the output of the last fiber optic.

3.6.27.11 Optical axis. The optical axis of the assembly is defined as the mean center line of that cylindrical portion of the assembly used to aline the assembly in the system housing.

3.6.27.12 Room temperature. Room temperature is defined as plus 23°C, ± 3°C.

3.6.27.13 Rise time. The rise time is the time required for the assembly to achieve 90 percent of its steady state output brightness after the voltage source is applied to the assembly.

3.6.27.14 Ambient useful steady state output brightness. The ambient useful steady state output brightness of the assembly is defined as the steady state output brightness with the operating potentials applied, the manual gain control set at maximum, and any specified steady state input illumination applied.

3.6.27.15 Failure definitions.

3.6.27.15.1 Equipment failure. The inability of the image intensifier assembly to perform its function as specified herein to include damage resulting from improper operation. Damage which results from abuse is not a failure. The need for adjustment of controls not normally available to the operator is considered to be a failure. The performance requirements are not restricted to those of Tables X and XI.

3.6.27.15.2 Pattern failure. The occurrence of two (2) or more failures of the same part in identical or equivalent application whose combined failure rate exceeds that predicted. When two (2) or more dependent failures of the same part occur whose combined failure rate exceeds that predicted and which are due to more than one independent failure, the dependent failures are relevant pattern failures. Unless a Government approved Reliability Math Model is available, failure rates found in MIL-HDBK-217 shall be used.

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3.6.27.15.3 Independent failure. A failure which will independently cause equipment performance outside of specified limits - one which occurs without being related to the failure of associated items.

3.6.27.15.4 Dependent failure. A failure of a part which is a direct result of an independent failure - one which is caused by the failure of an associated item(s). Dependent failures are not necessarily present when simultaneous failures occur.

3.6.27.16 Qualified product. A product that has successfully met all the requirements of an IPT or First Article witnessed by the CECOM, Concurrent Engineering Directorate or its designated representative. Should the supplier make substantial changes in the process, personnel, location, or equipment used to produce a qualified product, or have not delivered the qualified product to the Federal Government within 12 months of contract award, an IPT shall be performed to the extent necessary to verify continued compliance to specification requirements for all characteristics of the product affected by the changes.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3 and 5. The inspection set forth in this specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Parts and materials inspection. The supplier is responsible for certification that parts and materials used are manufactured, examined and tested in accordance with referenced specifications, standards, and drawings as specified herein.

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4.2 Classification of inspection. Inspection shall be classified as follows:

a. Initial Production Testing inspection (4.3).
b. Quality conformance inspection (4.5).
c. Inspection of packaging (4.7).

4.3 Initial Production Testing (IPT). Unless otherwise specified in the contract or purchase order (see 6.2.b), the IPT inspection shall be performed by the contractor.

4.3.1 Inspection. Each IPT assembly shall be examined in accordance with Table III. Presence of one or more defects shall be cause for rejection of that assembly.

TABLE III. Inspection.

Defects	Requirement paragraph
Components and materials missing or not as specified	3.5
Dimension not as specified	3.2
Identification or special marking missing or illegible	3.6.25
Workmanship not as specified	3.6.26

4.3.2 Tests. Following successful completion of the inspection specified in 4.3.1, each IPT assembly shall be subjected to all tests in Table IV. Failure of any test shall be cause for rejection of that assembly and may be cause for termination of IPT testing. IPT reliability sample shall consist of ten (10) assemblies.

TABLE IV. Initial Production Testing Inspection.

Inspection	Requirement paragraph	Test paragraph
Photocathode sensitivity	3.6.1	4.6.1
Burn-in (ESS)	3.6.2	4.6.2
Vibration	3.6.3	4.6.3
Mechanical shock - I	3.6.4.1	4.6.4.1
Mechanical shock - II	3.6.4.2	4.6.4.2
Equivalent background input	3.6.6	4.6.6
Luminance gain	3.6.7	4.6.7
Environmental temperature	3.6.5	4.6.5
Photocathode and screen quality	3.6.23	4.6.23
Fixed pattern noise	3.6.11	4.6.11
Center resolution	3.6.15.1	4.6.15.1
Peripheral resolution	3.6.15.3	4.6.16.3

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TABLE IV. Initial Production Testing Inspection - Continued.

Inspection	Requirement paragraph	Test paragraph
High light level center resolution	3.6.15.2	4.6.15.2
Modulation transfer function	3.6.16	4.6.16
Signal-to-noise ratio	3.6.10	4.6.10
Time dependent bright source	3.6.8	4.6.8
Electrical stability	3.6.20	4.6.20
Image alignment	3.6.13	4.6.13
Bright source protection	3.6.9	4.6.9
Output brightness uniformity	3.6.12	4.6.12
Magnification	3.6.18	4.6.18
Useful cathode diameter	3.6.17	4.6.17
Phosphor decay	3.6.21	4.6.21
Phosphor spectral response	3.6.22	4.6.22
Image shift	3.6.14	4.6.14
Shear distortion	3.6.19	4.6.19
Reliability	3.6.24	4.6.24

4.4 Inspection procedures. Tests shall be conducted in accordance with the test procedures specified herein. Unless otherwise specified, the following conditions shall apply:

a. The radiation source used in the tests shall be a tungsten filament lamp operated at a color temperature of 2856K, plus or minus 50K.

b. The photometer used for screen brightness shall be a Pritchard Model 1970 PR or equal.

c. The photometer used for screen brightness measurements shall be calibrated against a standard source which has a tungsten filament lamp, opal glass and filters as specified below.

(1) Tungsten filament lamp operated in conjunction with an opal glass such that the color temperature of the radiation emitted from the opal glass is 2856K, plus or minus 50K.

(2) Corning Spectral Filters, Nos. 3-71 and 4-67 or equal.

(3) Opal glass to produce a uniform, lambertian distribution.

(4) Output brightness to be 0.2 to 0.7 footlambert uniformly distributed over an output aperture of not less than 20 millimeters.

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d. The amount of radiation from the source incident on the photocathode for each test shall be the amount specified in that test. Tolerances on specified radiation levels shall be plus or minus 25 percent.

e. Meters used for monitoring lamp current and voltage shall have an accuracy of plus or minus 0.25 percent.

f. Tests shall be performed at room temperature (see 3.6.27.12).

g. Neutral density filters used in test equipment shall have transmission characteristics within 10 percent of the nominal filter transmission from 0.35 to 1.0 micrometer.

h. All tests on the assembly shall be performed with the input voltage at 2.50, ± 0.05 Vdc and at the maximum manual gain position.

i. All tests shall be performed with the assembly mounted in an approved grounded test housing.

j. Tolerances on applied nominal input voltages of 2.00 Vdc and 3.00Vdc shall be ± 0.05 Vdc.

k. Test chamber used for environmental temperature tests shall maintain the temperature within plus or minus 2°C (3.6°F) of the specified test temperature.

l. Test methods shall be in accordance with MIL-STD-202 or as specified herein.

4.5 Quality conformance inspection.

4.5.1 Inspection lot. For purposes of inspection, a lot shall be defined by the contractor.

4.5.2 Tests.

4.5.2.1 Group A inspection. Each intensifier shall be subjected to the inspection in Table III and the tests in Table V. The tests in Table V shall be performed in the sequence listed. Presence of one or more defects or failure of any test shall be cause for rejection of that intensifier.

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TABLE V. Group A inspection.

Inspection	Requirement paragraph	Test paragraph
Photocathode sensitivity	3.6.1	4.6.1
Burn-in (ESS)	3.6.2	4.6.2
Equivalent background input	3.6.6	4.6.6
Luminance gain	3.6.7	4.6.7
Photocathode and screen quality	3.6.23	4.6.23
Fixed pattern noise	3.6.11	4.6.11
Center resolution	3.6.15.1	4.6.15.1

4.5.2.2 Group B inspection. Intensifiers passing Group A inspections shall be formed into lots in accordance with 4.5.1 and shall be subjected to Group B inspection as defined in Table VI. Unless otherwise stated, sampling shall be in accordance with the contract or purchase order (see 6.2.d). Tests within each subgroup, may be performed in any order.

4.5.2.2.1 Subgroup 4. Testing for Subgroup IV shall only be performed in the event a new batch of phosphor is utilized. One (1) assembly shall be selected from the first ten (10) assemblies produced with the new batch of phosphor. This single assembly shall be subjected to the tests of Subgroup IV. In the event of failure of either or both of the tests in Subgroup IV, all assemblies containing phosphor from the lot shall be rejected.

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TABLE VI. Group B inspection.

Inspection	Requirement paragraph	Test paragraph
Subgroup 1		
Output brightness uniformity	3.6.12	4.6.12
MTF	3.6.16	4.6.16
Signal-to-noise-ratio	3.6.10	4.6.10
Subgroup 2		
Electrical stability	3.6.20	4.6.20
Image alignment	3.6.13	4.6.13
Magnification	3.6.18	4.6.18
Subgroup 3		
Time dependent bright source	3.6.8	4.6.8
Bright source protection	3.6.9	4.6.9
Image shift	3.6.14	4.6.14
High light level center resolution	3.6.15.2	4.6.15.2
Shear distortion	3.6.19	4.6.19
Useful cathode diameter	3.6.17	4.6.17
Peripheral resolution	3.6.15.3	4.6.15.3
Subgroup 4		
Phosphor decay	3.6.21	4.6.21
Phosphor spectral response	3.6.22	4.6.22

4.5.2.3 Group C inspection. This inspection shall consist of the tests specified in Table VII and shall be performed on units that have been subjected to and met Group A inspection. Sampling plans shall be in accordance with the contract or purchase order (see 6.2.d).

TABLE VII. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
Vibration	3.6.3	4.6.3
Mechanical shock - I	3.6.4.1	4.6.4.1
Mechanical shock - II	3.6.4.2	4.6.4.2
Environmental temperature	3.6.5	3.6.5

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4.5.2.4 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order (see 6.2.c).

4.5.2.5 Reinspection of conforming Group C samples. Unless otherwise specified, the samples which have been subjected to and passed Group C inspection shall be accepted on the contract provided all damages are repaired. Samples which were subjected to Humidity shall be baked a minimum of 55°C and 400mm Hg maximum for a minimum of 24 hours prior to use.

4.5.2.6 Group D inspection. This inspection shall consist of the tests specified in Table VIII. For IPT, ten assemblies shall be selected at random for the Group D inspection. For subsequent quality conformance, Group D inspections, 2 assemblies shall be selected at random from the first lot of each months' production from the assemblies which have passed Group A inspection. If the production quantity exceeds 200 assemblies in a given month, then an additional sample of two will be taken. An additional sample of twenty (20) assemblies shall be randomly selected from the same lot that the Group D samples were selected from, identified as verification samples, and held until the Group D tests have been completed. The twenty samples shall not be selected from the second, sixth, and tenth months first lot.

TABLE VIII. Group D inspection.

Inspection	Requirement paragraph	Test paragraph
Reliability	3.6.24	4.6.24

4.5.2.7 Group D failures. Action required relative to Group D failures shall be as specified in the contract or purchase order (see 6.2.c).

4.5.2.8 Reinspection of assemblies conforming to Group D inspection. Unless otherwise specified, assemblies which have been subjected to and successfully completed Group D inspection may be accepted on the contract provided each assembly meets all of the following:

a. All damages are repaired.

b. If the luminance gain of the assembly at post MTTF testing is 20,000 or greater at the light level of 5×10^{-6} fc, the assembly shall be refurbished as required. Such assemblies, after refurbishment and repair of all damage, shall be subjected to and pass the Group A inspection of 4.5.2.1, except cathode sensitivity (4.6.1) and Burn-in (ESS) (4.6.2), before they are accepted in the contract.

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4.5.2.9 Disposition of Group D reliability inspection assemblies. Upon completion of the IPT and the second, sixth and tenth Group D reliability requirements, samples shall be subjected to additional reliability testing (for information only) until samples reach end of life (inability to meet the requirements specified in the reliability failure definition) or 4000 hours, whichever comes first. Samples and data shall then be shipped to CECOM, Concurrent Engineering Directorate, ATTN: AMSEL-ED-IN, Fort Monmouth, NJ 07703. The balance of the Group D reliability samples shall be refurbished, pass Group A inspection and shipped per 4.5.2.8.

4.6 Test methods.

4.6.1 Photocathode sensitivity. This test shall be performed on the assembly prior to final installation of the power supply and boot. A dc voltage sufficient to collect the full cathode current shall be applied. Corrections for leakage and dark current shall be made. Illuminate a 19.5, ± 0.5 millimeter area centered on the photocathode with .01 to .02 lumen of 2856K radiation. The photocathode current in microamperes divided by the actual input (lumens) is the 2856 K photocathode sensitivity. With the dc voltage as above, insert a .80 micrometer filter between the phtocathode and the light source. Illuminate a 19.5, ± 0.5 millimeter area with a 1×10^{-5} to 1×10^{-4} watt of .80 micrometer radiation. Measure the total assembly current and subtract the leakage and dark current determined above. This photocurrent (in microamperes) divided by the actual input radiation (in watts) is the photocathode radiant sensitivity at .80 micrometer. Remove the .80 micrometer filter and insert the 0.85 micrometer filter between the photocathode and the light source. The 19.5, ± 0.5 millimeter area on the photocathode shall be illuminated with 1×10^{-5} to 1×10^{-4} watt of 0.85 micrometer radiation. Measure the total assembly current and subtract leakage and dark current. This photocathode current divided by the actual input radiation (in watts) is the photocathode sensitivity at 0.85 micrometer. The 0.80 and 0.85 micrometer filters shall have the following characteristics:

a. Far infrared blocking out to 4 micrometers and ultraviolet blocking down to 0.3 micrometer.
b. Peak wavelength of 0.8000, plus or minus .0025 micrometer, and 0.8500, plus or minus 0.0010 micrometer.
c. Bandwidth at the 10 percent points of 0.0125 micrometer, plus or minus 0.0015 micrometer.
d. Minimum peak transmission of 50 percent.

Failure to meet the requirements specified in 3.6.1 shall constitute failure of this test.

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4.6.2 Burn-in/Environmental Stress Screening (ESS). With the operating voltage applied, illuminate the photocathode at least once each 12 minute cycle, with 1 to 5 x 10^{-3} footcandle for not less than 3 seconds and 5 to 10 footcandles for not less than 5 seconds (see Figure 2). There shall be no illumination on the photocathode during the remainder of the 12 minute cycle. Failure to meet the requirements of 3.6.2 shall constitute failure of this test.

4.6.2.1 High temperature stress test. After pretests of photocathode sensitivity and photo response the assembly shall be placed in a chamber at +52°C with the photocathode terminal at -4000Vdc and the ANODE cone terminal at ground potential. There shall be no illumination. After 48 hours minimum time under stress the assemblies shall be removed and after a 4 hour maximum rest shall be post tested. Failure to meet the requirements of 3.6.2.1 shall constitute failure of this test.

4.6.2.2 High input illumination stress test.

Pre Test: Measure Photocathode Sensitivity.

Test: Apply operating potentials sufficient to produce 50K luminance gain. Illuminate photocathode to generate photocathode current equivalent to that generated by 1 x 10^{-3} f.c. white light. (Note: illumination need not be white light). Maintain operating potentials and illumination for 48 hours.

Post Test: Measure Photocathode Sensitivity and photo response. Tube Module must meet the requirements of 3.6.2.2 and 3.6.1. Failure to meet the requirements of 3.6.2.2 and 3.6.1 shall constitute failure of this test.

4.6.3 Vibration. The assembly shall be subjected to Method 201A of MIL-STD-202. The operating potential shall be applied during vibration testing. Tolerances on specified frequencies shall be plus or minus 1 percent. Prior to beginning vibration testing, the assemblies shall be visually inspected for damage and missing components. No assembly shall be vibrated which shows evidence of damage or missing components. Mount the assemblies rigidly, singly or in groups. Subject the assemblies to simple harmonic motion applied in planes parallel and perpendicular to the optical axis. In 30 minutes, vary the frequency from 10 Hz to 3500 Hz and return to 10 Hz in a logarithmic sweep. Perform the frequency sweep two times in each plane. Failure to meet the requirements of paragraph 3.6.3, 3.6.6, 3.6.7, 3.6.15.1, and 3.6.24 shall constitute failure of this test.

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4.6.4 Mechanical shock.

4.6.4.1 Mechanical shock I. This test is to be conducted in a darkened room with no light incident on the photocathode of the assembly and the operating potential applied. The assembly shall be rigidly mounted in approved test fixture. Subject the assembly to the shock impulse specified in 3.6.4.1. The image screen shall be observed with the unaided eye during the application of the impacts. There shall be no more than two flashes in each axis. Test shall be monitored with an accelerometer mounted on the test fixture near the assembly. Pulse shape and amplitudes shall be monitored. Failure to meet the requirements of 3.6.4.1, 3.6.6, 3.6.7, 3.6.15.1, and 3.6.23 shall constitute failure of this test.

4.6.4.2 Mechanical shock II. This test is to be conducted in the same manner as described in 4.6.4.1 Subject the assembly to the shock impulse specified in 3.6.4.2. There shall be no more than two flashes in each axis. Failure to meet the requirements of 3.6.4.2, 3.6.6, 3.6.7, 3.6.15.1, and 3.6.23 shall constitute failure of this test.

4.6.5 Environmental. Place the assembly(s) in a test chamber at room temperature. Perform the room temperature tests specified in Table IX. The test procedure for EBI and Gain shall be the same as specified in 4.6.6 and 4.6.7. Remove the operating potential from the assembly. Raise the temperature of the test chamber to plus 68°C and hold at this temperature for a minimum of one (1) hour. At the end of the one (1) hour stabilizing period, lower the test chamber temperature to plus 52°C and adjust the relative humidity to 95 percent. At the end of a four (4) hour period with the temperature at plus 52°C, reduce the relative humidity to less than 25 percent. Maintain this condition for one (1) hour. At the end of the hour subject the assemblies to the high temperature performance tests specified in Table IX. The rise time shall be recorded on a X-Y recorder or equivalent. The overshoot recorded during rise time shall be less than 40 percent of steady state output brightness. The operational stability of the assembly shall be verified by viewing the output signal from the photometer with a storage oscilloscope. Set the sweep time for not less than 30 seconds across the screen. Adjust the vertical sensitivity to obtain a minimum of 80 percent deflection of the total oscilloscope screen from a zero reference. View the screen for fluctuation. All fluctuation shall be random. Fluctuation, at random, greater than plus or minus 10 percent of steady state shall constitute failure of this test. After high temperature tests are completed, remove the operating potential to the assembly and thermal shock the units from plus 52°C, to room temperature in 5 minutes or less. Place the assemblies in test chamber and reduce the temperature to minus 62°C and hold for one (1) hour. At the end of the one (1) hour stabilizing period, raise the chamber temperature to minus 54°C. Hold at this temperature for one (1) hour. At the end of the one (1) hour soak, subject the assemblies to the cold temperature performance tests specified in Table IX. The test procedures for rise time, gain, and operational stability shall be the same as those used in high temperature.

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At the conclusion of cold temperature tests, and with operating potential removed, thermal shock the assemblies from minus 54°C to room temperature in 5 minutes or less. Hold the units at room temperature for at least one (1) hour but not more than 16 hours. At the end of the holding period, subject the units to the room temperature tests specified in Table IX. The test procedures and conditions shall be identical to the plus 23°C test prior to high temperature test. Place the assemblies and a BA-3058/U or BA-5567/U voltage source in the test chamber. Lower the chamber temperature to minus 17.8°C and hold for one-half hour. Perform the gain test at minus 17.8°C. Raise chamber temperature to room temperature and remove assemblies. The 95 percent humidity test at 52°C may be run separately using the times indicated in Figure 3. Failure to meet the requirements of 3.6.5, 3.6.6, 3.6.7, 3.6.15.1, and 3.6.23 shall constitute failure of this test .

High temperature gain and EBI are defined as:

$$G_{htjl} = \frac{B_{hjl} - B_{hol}}{B_{rjl} - B_{rol}} \text{ X ROOM TEMPERATURE GAIN}$$

$$EBI = \frac{B_{hol}}{B_{rol}} \text{ X ROOM TEMP. BRIGHTNESS (W/O ILLUMINATION)} \frac{(1.08 \times 10^{-3})}{G_{htjl}}$$

Low temperature gain and EBI are defined as:

$$G_{ltjl} = \frac{B_{ljl} - B_{lol}}{B_{rjl} - \text{Broi}} \text{ X ROOM TEMPERATURE GAIN}$$

$$EBI = \frac{B_{lol}}{B_{rol}} \text{ X ROOM TEMP. BRIGHTNESS (W/O ILLUMINATION)} \frac{(1.08 \times 10^{-3})}{G_{ltjl}}$$

Where:

B_{hjl} - Screen brightness at high temperature with applicable input levels.

B_{hol} - Screen brightness at high temperature with no light input.

B_{rjl} - Screen brightness at room temperature with applicable input light levels.

B_{rol} - Screen brightness at room temperature with no light input.

B_{lol} - Screen brightness at low temperature with no light input.

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B_{III} - Screen brightness at low temperature with applicable input levels.

4.6.6 Equivalent background input. With the operating potential applied to the assembly and no radiation incident on the photocathode, hold for a stabilizing period of not more than 15 minutes. At the end of the stabilizing period with no radiation incident on the photocathode measure the screen brightness (footlamberts) with a photometer. Observe the readings for 30 seconds and record the maximum and minimum readings, R1. Illuminate the faceplate with 2 x 10^{-11} lumen/cm². Observe the reading for 30 seconds and record maximum and minimum readings, R2. The angle of incident flux shall be not greater than 2°. The photometer shall be positioned such that the acceptance angle covers 19mm, plus or minus 0.5mm diameter area centered on the screen. Average the maximum and minimum dark readings R1 and the maximum and minimum illuminated readings R2 and determine equivalent background input (EBI) by the following formula:

$$EBI = \frac{R_1}{R_2 - R_1} \text{ X ACTUAL INPUT ILLUMINATION}$$

Equivalent background input greater than specified in 3.6.6 shall constitute failure of this test.

4.6.6.1 EBI-Alternate 1. An alternate method for measuring EBI which may be used in lieu of that specified in 4.6.6 is as follows:

$$EBI = \frac{I1 - I_o}{I_2 - I1} \text{ X } (2 \text{ X } 10^{-11} \text{ LUMENS/cm}^2)$$

Where:

I_o -Dark current of photomultiplier tube.
I_1 -Photomultiplier current due to brightness of intensifier with no incident radiation.
I2 -Photomultiplier current due to brightness of intensifier when illuminated with 2 x 10^{-11} lumen/cm².

Equivalent background input greater than that specified in 3.6.6 shall constitute failure of this test.

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4.6.6.2 EBI-Alternate 2. An additional alternate method of measuring EBI may be used in lieu of that specified in 4.6.6 and 4.6.6.1 is as follows: With the operating potential applied to the intensifier and no radiation incident on the photocathode hold for a stabilizing period of not more than 15 minutes. At the end of the stabilizing period with no radiation incident on the photocathode measure the screen brightness (footlamberts) with the photometer. Divide this reading by the luminance gain and multiply the result by 1.08×10^{-3}. An EBI greater than that specified in 3.6.6 shall constitute failure of this test.

4.6.7 Luminance gain. Uniformly illuminate the photocathode in accordance with each of the conditions specified in Table I. Apply the operating potential for the appropriate manual gain setting to the assembly. Measure the input current and measure the image screen luminance with the photometer. The acceptance angle of the photometer shall be 2° or less. Position the photometer so that the acceptance angle subtends a 19.5, ± 0.5 millimeter diameter spot concentric with the useful screen diameter. Luminance gain is determined by dividing the screen luminance (footlamberts) by the actual illumination (footcandles). Luminance gain and input current less than or greater than that specified in 3.6.7, Figures 4 and 5 shall constitute failure of this test.

4.6.8 Time dependent bright source.

4.6.8.1 Recovery time (Figure 6). Using a light source and an input steady state ambient illumination of 1 to 5×10^{-5} footcandles with the assembly at operating potential and maximum manual gain setting, apply the pulsed input illumination to the assembly in accordance with profile (a) Figure 6. Record the recovery time from the end of the applied pulse. Failure to meet the minimum requirements of 3.6.8.1 shall constitute failure of this test.

4.6.8.2 Recovery time (Figure 7). Using the conditions set forth in 4.6.8.1, apply the periodic pulsed input illumination to the assembly in accordance with profile (b) Figure 7. Apply this profile through 3 complete and continuous cycles in such a manner that the third cycle ends on a simulated tracer pulse. Determine the recovery time from the end of this last tracer pulse. Failure to meet the minimum requirements set forth in 3.6.8.2 shall constitute failure of this test.

4.6.8.3 Resolution (Figure 7). Butt or focus a standard black on clear Resolving Power Test Target USAF 1951 onto the photocathode of the assembly. Apply concurrently with the resolving power test target the periodic pulsed input illumination of profile (b) Figure 7. View the output screen with a 10X viewing system. Failure to meet the requirements of 3.6.8.3 shall constitute failure of this test.

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4.6.9. Bright source protection. With operating potential applied illuminate the photocathode with a spot of light not greater than 1.0 square millimeter in area, having an intensity of not less than 50 millilumens (2856K not required). The 50 millilumens is referenced to a 240 uA/lumen cathode. The lumen input may be decreased for high sensitivity cathodes such that the cathode current is the same as the 50 millilumen/240 uA cathode combination. Apply the light spot on the photocathode for 20 seconds, ± 1 second. Failure to meet the requirements of 3.6.9 shall constitute failure of this test. NOTE: The flicker inherently associated with the assembly when the bright source is applied and removed from the assembly shall not be cause for failure.

4.6.10 Signal-to-Noise ratio. Image a circular spot no larger than 0.200 millimeter in diameter onto the photocathode of the assembly. The circular spot shall uniformly illuminate the photocathode at a level not greater than 1.2×10^{-6} footcandle. Focus the signal which is emergent from the assembly on a pinhole 0.200, ± 0.050 millimeter in diameter. Align the pinhole to obtain a maximum signal. Measure the light passing through the pinhole with a low dark current photomultiplier tube, EMR Model 541E or equivalent. Pass the signal to a Melpar S/N measurement set Model No. R458997 or equivalent and measure the dc content and the rms value of the signal over an electronic bandwidth of 10 Hertz. The signal-to-noise (S/N) is the ratio of the dc signal to the rms noise.

$$S/N = \frac{S_o - S_{bkd}}{K\,(N_o^2 - N_{bkd}^2)^{1/2}}$$

WHERE:

S - dc signal

N - rms noise

S_o - Signal output from a Melpar Type S/N test set or equivalent.

S_{bkd} - Background signal when the light input to the assembly is closed.

N_{bkd} - Background noise as measured by a Melpar Type S/N test set rms meter or equivalent, when the light input to the assembly is closed.

K - A correction factor to obtain a signal-to-noise ratio over an equivalent bandwidth of 10 hertz independent of the frequency response of the assembly. K = 1.30 for a phosphor having temporal decay characteristics as specified in 3.6.21.

Failure to meet the requirements of 3.6.10 shall constitute failure of this test.

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4.6.11 Fixed pattern noise. With the operating potential applied to the assembly, uniformly illuminate the entire photocathode at a level not less than 1 x 10^{-4} footcandle nor greater than 5 x 10^{-4} footcandle and observe the phosphor screen at minimum and maximum gain setting, with a 10 power eyepiece, for graininess, multi-to-multi pattern variation, mottling, multiboundary noise (webbing) and shading with lines of demarcation. Repeat the tests with an input illumination of 1 x 10^{-3} footcandle to 5 x 10^{-3} footcandle. If multi-to-multi pattern variations or multi-boundary pattern noise are observed as either maximum for minimum gain setting for either light level perform the following tests at each input illumination level for which the fixed pattern noise was observed.

4.6.11.1 Multi-to-multi. Choose an area of the image screen where the multi-to-multi pattern appears most noticeable. Search this area for the most contrasting adjacent multibundles and measure each multibundle brightness using an aperture of effective diameter equal to 1/3 of the dimension of a multibundle diameter (as measured from flat to flat for hexagonal multies). Failure to meet the requirements of 3.6.11.1 shall constitute failure of this test.

4.6.11.2 Multi-boundary. Choose an area of the image screen where the multi-boundary pattern noise is most noticeable. Scan 3 multies and corresponding multi-boundaries in this area of the screen with a photometering aperture of 25 micrometers effective diameter. Establish a mean value of intensity using the average values of the multies. Establish the average value of the 4 corresponding multi-boundaries and note their polarity. This average value must not deviate from the mean value by more than that which is allowed in 3.6.11.2.

4.6.12 Output brightness uniformity. When viewed with a 10X magnifier, the background shading shall be uniformly graded with no distinct line of demarcation between the light and dark areas with the exception of the dark area inherent in inverter assemblies. This dark area shall not fall within a circle 0.6 inch in diameter concentric with the optical axis. Output brightness uniformity shall be verified as follows:

(1) Uniformly illuminate the photocathode of the assembly at a level of 1 to 5 x 10^{-4} footcandle.

(2) Scan completely across the assembly output image screen with a photometer with an aperture such that its effective size on the screen of the assembly is 2.5mm. Record the photometer output on an X-Y recorder or equivalent.

(3) Rotate the assembly under test 45° about its optical axis and repeat step (2).

(4) Rotate the assembly an additional 45° about its optical axis in the same direction as in (3) above and repeat step (2).

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(5) Repeat step (4). Calculate the uniformity ratio by using the maximum and minimum values of the combines scans. Failure to meet the requirements of 3.6.12 shall constitute failure of this test.

4.6.13 Image alignment. With the operating potential applied, focus or butt the test pattern described in Figure 9 or 10 or a spot of light 0.127 mm in diameter on the photocathode. The photocathode shall be illuminated to provide a high contrast image of the test pattern. Observe the image of the test pattern formed on the screen of the intensifier with a 10X or greater measuring microscope containing a reticle of 0.040 inch apparent diameter. The microscope shall be aligned with the optical axis of the intensifier such that the center of the 0.040 inch diameter reticle falls on the optical axis. The center of the test pattern shall fall within the 0.040 inch diameter circle. Failure to meet the requirements of 3.6.13 shall constitute failure of this test.

4.6.14 Image shift. Determine image shift with same equipment used in 4.6.13. At the conclusion of the image alignment test position the center of the reticle of the measuring microscope on the center of the test reticle and remove the operating potential from the assembly. After the assembly is completely discharged, apply the operating potential and observe the test reticle for 30 seconds. Should the test reticle be displaced from its original position prior to removing the operating potential) measure the displacement. Image shift greater than that specified in 3.6.14 shall constitute failure of this test.

4.6.15 Resolution.

4.6.15.1 Center resolution. Perform this test using:

(a) A radiation source in accordance with 4.4.

(b) A focusing system having an f-number not greater than 1.2 to project the test pattern.

(c) A resolving power target having black bars on a clear background (Resolving Power Test Target USAF 1951), Figure 10.

(d) A 10X or greater microscope. With a photocathode input illumination level to the assembly 1 x 10^{-4} to 1 x 10^{-3} footcandle uniformly distributed over the useful photocathode, focus or butt the resolving power target onto the photocathode in such a manner that the center of the target is aligned with the optical axis of the assembly. The image of the resolving power target formed on the screen of the assembly shall be observed for limiting resolution (see 3.6.27.7). Resolution less than that specified in 3.6.15 shall constitute failure of this test.

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4.6.15.2 High light level center resolution. Repeat this test with an input photocathode illumination of not less than 0.8 nor greater than 5 footcandle. Resolution less than that specified in 3.6.15.1 shall constitute failure of this test.

4.6.15.3 Peripheral resolution. Determine the peripheral resolution using the same equipment as in 4.6.15. The resolving power target shall be focused or butted on the photocathode of the assembly in such a manner that elements of the target are positioned 10.0mm, ± 0.5 millimeters from the optical center of the photocathode. Repeat this test at three positions 90°, ± 5° from the original test position. Resolution less that specified in 3.6.15.3 shall constitute failure of this test.

4.6.16 Modulation transfer functions (MTF). Equipment used to measure modulation transfer function must satisfy the following or equivalent subject to Government approval:

a. A sine wave analyzer capable of direct readout at (but not limited) to spatial frequencies within one line pair of the following: 2.5 line pairs per millimeter, 7.5 line pairs per millimeter, and 15 line pairs per millimeter. Its spatial frequency accuracy must be ± 3 percent, and accuracy of MTF at each frequency must be ± 2 percent of the value.

b. The analyzing slit must be 10 micrometers or less in width by 1mm or more in length.

c. The limiting aperture in the plane of the test pattern shall be a minimum of 2.5mm in width referred to the phosphor screen of the tube assembly.

d. The test system calibration frequency shall be 0.2 lp/mm or less.

e. The test system MTF shall be greater than 95 percent at 2.5 lp/mm.

f. Input illumination measured at the assembly photocathode shall be less than 2×10^{-3} fc.

The zero MTF level shall be determined by blocking the light from the phosphor screen while the system is running at calibration frequency. When unblocked, the calibration reading is normalized to 100 percent MTF. Assembly MTF is arrived at by dividing MTF of the measuring system with tube in place by MTF of the measuring system without the tube. Failure to meet the requirements of 3.6.16 shall constitute failure of this test.

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4.6.17 Useful cathode diameter. Perform this test with the same equipment used in 4.6.15. With the operating potential applied to the assembly, focus the test pattern on the optical axis of the photocathode at unity magnification and adjust the input radiation level for best image contrast. The image screen shall be observed with a 10X or greater microscope. Useful photocathode diameter is determined by the number of millimeter graduations visible on the image screen of the assembly and shall be determined in both vertical and horizontal directions. Failure to meet the requirements of 3.6.17 shall constitute failure of this test.

4.6.18 Magnification. The magnification of the assembly shall be measured at the center and at the radial distance of 1 millimeter, 5 millimeters, 10 millimeters and 11.5 millimeters. Perform this test with the same equipment used in 4.6.13 or 4.6.17. With the operating potential applied to the assembly, focus the center of the test pattern onto the optical axis of the photocathode and adjust the input radiation level for best image contract. The test pattern shall be the same as the pattern described in Figure 10. Measure separation in millimeters of the graduations (Y) on the image screen of the assembly and the distances from the center of the test pattern to the left and right 1.0 millimeter (Y_1), 5 millimeters ($Y_{5.0}$), 10 millimeters ($Y_{10.0}$), and 11.5 millimeters ($Y_{11.5}$) graduations.

Compute the magnification as follows:

a. Center magnification $= \frac{Y_c}{2.000} M_c$

b. Peripheral magnification (at 11.5 millimeters from the optical axis on the photocathode).

$= \frac{Y_{11.5}}{11.500} = M_p$

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4.6.19 Shear distortion. Inspect the total image screen of the assembly for shear distortion. If shear distortion is observable at any point, using a 10X or greater viewing system and the line test pattern shown in Figure 11, project a line image in such a manner that is intersects the shear distortion. Using a measuring microscope whose direction of scan is perpendicular to the direction of the line image, measure the shear distortion. Shear distortion exceeding that specified in 3.6.19 shall constitute failure of this test.

4.6.20 Electrical stability. With the operating potential applied to the assembly and no radiation incident upon the photocathode, observe the assembly for damage. While measuring the input current, probe the entire open surface of the assembly with a grounded 1/4 inch diameter probe. The probe shall be applied within 0.050 inch, plus or minus 0.005 inch of the fiber optic faceplate of the assembly. Failure to meet the requirements of 3.6.20 shall constitute failure of this test.

4.6.21 Phosphor decay. The phosphor decay characteristics of the assembly shall be determined by uniformly illuminating within 25 percent, the entire photocathode with a pulsed light (2856K) color temperature is not required) whose dark to light time is less than one millisecond, whose light time is 125 milliseconds, ± 10 milliseconds, and whose light to dark time is less than 100 milliseconds. The dark light level shall be less than 4×10^{-7} footcandles. The instrumentation for the analysis of the decay time shall consist of a signal (pickup) system and a trigger system. The signal system shall detect the output light from the phosphor. The resulting signal shall go to a signal analyzing device. The response time of the entire signal system shall be less than 300 microseconds. The trigger system shall provide a means to obtain an electronic pulse directly related to the input light pulse. This trigger pulse must be capable or triggering the signal analyzing device with a repeatability of plus or minus 300 microseconds in terms of trigger time versus actual start or decay. The signal analyzing device must be capable of looking at the decay in several section so that the value read at a particular time from 10 milliseconds is not less than 20 percent of full scale or equivalent. Full scale accuracy shall be ± 0.5 percent. Values shall be taken at 10 milliseconds, 100 milliseconds, 500 milliseconds, and 1 second. Failure to meet the requirements of 3.6.21 shall constitute failure of this test.

4.6.22 Phosphor spectral response. This test shall be performed using a scanning spectroradiometer, Gamma Model 3000 or equal. Illuminate the full photocathode with an input illumination sufficient to provide an output brightness that can be detected by the spectroradiometer. Scan the phosphor screen with the spectroradiometer and determine the peak response wavelength, the 10 percent points of the spectral response, the bandwidth between the 10 percent points, and the relative response at the 6500 angstrom wavelength. Failure to meet the requirements of 3.6.22 shall constitute failure of this test.

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4.6.23 Photocathode and screen quality.

4.6.23.1 High light quality. With the operating potential applied and radiation incident on the photocathode of the assembly, observe the image screen with a 10X or greater microscope. Adjust the light level to obtain best spot contrast between bright and dark spots and observe the image screen for graininess, or spots which exceed the size or quantity specified in Table II. Failure to meet the requirements of 3.6.23 shall constitute failure of this test.

4.6.23.2 Low light quality. With the operating potential applied and an input radiation level of 5×10^{-3} footcandle uniformly applied to the full photocathode, hold for a minimum of one minute. Reduce the input radiation level to not greater than 2×10^{-7} footcandle and observe the image screen with the 10X or greater microscope for bright spots, ion spots, and field emission. Failure to meet the requirements specified in 3.6.23 shall constitute failure of this test.

4.6.24 Reliability. The reliability test shall be conducted in accordance with Table XII and as specified herein. ON-OFF cycling shall consist of 55 minutes ON time followed by 5 minutes OFF. During operation each assembly shall be illuminated with not less than 1×10^{-5} footcandles (fc) with the manual gain set at maximum. Once during each one hour cycle the illumination of each assembly shall be increased to 1×10^{-3} fc for 5 seconds, reduced to 1×10^{-5} fc for at least 5 minutes, and then increased to from 10 to 20 fc for 3 seconds. Sufficient instrumentation shall be provided to insure immediate recognition of a catastrophic failure. This shall include continuous recording of input current to each assembly, as well as detecting a change in the relative output brightness of each assembly. Once every 200 ± 50 hours of operation, each assembly on test shall be subjected to all of the tests in Table X. Reliability test assemblies shall be removed from test conditions for not more than 15 hours for measurement of assembly parameters. At other times, assemblies shall be cycled continuously on a 24 hour basis. Upon failure of any test, the failed assembly shall be removed from testing for failure analysis. Failed units will not be returned to test unless for information only. For testing in accordance with Table X, the acceptable minimum limit for EBI shall be 5×10^{-11} lumen/cm^2. Luminance gain shall be measured at 5×10^{-6} fc only. Acceptable limit for luminance gain shall be 20,000 at this light level. Failure to meet the requirements of 3.6.24 shall constitute failure of this test. For failure definitions see 3.6.27.15. Failures of the power supply shall be considered relevant. The requirements of MIL-STD-781 shall apply.

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TABLE IX. Environmental tests.

Temperature	Input-light (fc)	EBI 2.7 Vdc pot-max	Gain & input current 2.0 Vdc pot-max	Gain & input current 3.0 Vdc pot-max	Gain & input current 2.0 Vdc pot-min	Gain & input current 3.0 Vdc pot-min	Rise time 2.0 Vdc pot-max	Rise time 2.7 Vdc pot-max	Operational stability 2.0 Vdc pot-max	Battery rise time and gain
+23°C	0	*								
	1 to 5 x 10^{-6}		*	(*)	(*)	(*)				
	0.5 to 1 x 10^{-3}		*	(*)	(*)	(*)				
	0.01 to 0.03		(*)	(*)	(*)	(*)				
	10 to 20		*	(*)	(*)	(*)				
+52°C	0	*								
	1 to 5 x 10^{-6}		*	(*)	(*)	(*)	*	(*)		
	0.5 to 1 x 10^{-3}		*	(*)	(*)	(*)				
	0.01 to 0.03		(*)	(*)	(*)	(*)				
	0.05 to 0.1								*	
	10 to 20		*	(*)	(*)	(*)				
-54°C	0	*								
	1 to 5 x 10^{-6}		*	(*)	(*)	(*)	*	(*)		
	0.5 to 1 x 10^{-3}		*	(*)	(*)	(*)				
	0.01 to 0.03		(*)	(*)	(*)	(*)				
	0.05 to 0.1								(*)	
	10 to 20		*	(*)	(*)	(*)				
-17.8°C	1 to 5 x 10^{-6}		(*)	(*)						(*)

(*) Initial Production Test only.

NOTE: Input current readings not required at input light levels of 0.01 to 0.03 fc and 10 to 20 fc.

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TABLE X. Performance verification tests.

Test	Requirement paragraph	Test paragraph
Equivalent background input	4.6.24	4.6.6
Luminance gain	4.6.24	4.6.7
Signal-to-noise ratio	3.6.10	4.6.10
Damage	3.6.27.1	3.6.27.1

TAbLE XI. Final reliability inspection.

Test	Requirement paragraph	Test paragraph
Cathode and screen quality	3.6.23	4.6.23
Fixed pattern noise	3.6.11	4.6.11
Center resolution	3.6.15.1	4.6.15.1

4.6.24.1 Initial production test reliability inspection. For IPT reliability inspection, ten (10) assemblies shall be subjected to the MTTF test specified herein. The accept/reject determination shall be made on either the accumulation of 2500 hours operating time on each of the ten assemblies or the occurrence of four failures. Failure of four or more assembly failing to pass the tests of Table X and/or Table XI shall constitute failure of the equipment to meet the reliability inspection requirement. Three or less assembly failures shall constitute acceptance of the reliability inspection requirement.

4.6.24.2 Quality conformance reliability test inspection. For quality conformance testing, four assemblies shall be subjected to the MTTF test specified herein. The accept/reject determination shall be made on either the accumulation of 2500 hours on each of the four assemblies or the occurrence of two assembly failures. Failure of two or more assemblies to pass the tests of Table X and/or XI shall constitute failure of the equipment to meet the reliability inspection and be cause for rejection of the represented lots. One or less assembly failures shall constitute acceptance of the reliability inspection requirement.

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TABLE XII. Reliability test environmental conditions.

Parameter	
Temperature	50°C, ± 5°C
Temperature cycling	Not applicable
Vibration	2.2 g, ± 10% peak acceleration any nonresonant frequency between 20 and 60 Hz, measured at the mounting points on the equipment. The duration of vibration shall be at least 10 minutes during each period of equipment ON time.

4.7 Packaging inspection. Packaging and packing shall be inspected to assure compliance with section 5 Packaging requirements.

5. PACKAGING

5.1 Packaging requirements. The packaging requirements for the desired level(s) of protection shall be as specified by the acquisition activity (see 6.2.g).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1. Intended use. The image intensifier assembly covered by this specification is intended for use in an electro-optical viewing device to intensify low light level such that a visible image is presented for viewing and sighting purposes.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Quantity and schedule for IPT testing for all line items and disposition of IPT samples.

c. Necessary actions by the contractor in the event of a Group C failure (see 4.5.2.4) and/or a Group D failure (see 4.5.2.7).

d. Sampling plan for Group B and Group C testing. For guidance, unless otherwise specified, sampling shall be per the requirement of MIL-STD-105.

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e. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2)

f. MIL-STD-810 is used for all environmental tests of section 4.

g. Packaging requirements.

6.3 Qualification. With respect to products requiring qualification, unless otherwise specified by the contract or purchase order, if a product is not qualified at time of award, the contract shall require qualification prior to first delivery. The activity responsible for approving qualified products is the CECOM, Concurrent Engineering Directorate, ATTN: AMSEL-ED-IN, Fort Monmouth, NJ 07703. Information pertaining to qualification of products may be obtained from the above activity.

6.4 Definitions. See 3.6.27.

6.5 International standardization agreement. Certain provisions of this specification are the subject of international standardization agreement (QSTAG 468). When amendment, revision, or cancellation of this specification is proposed that will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels, including departmental standardization offices, to change the agreement or make other appropriate accommodations.

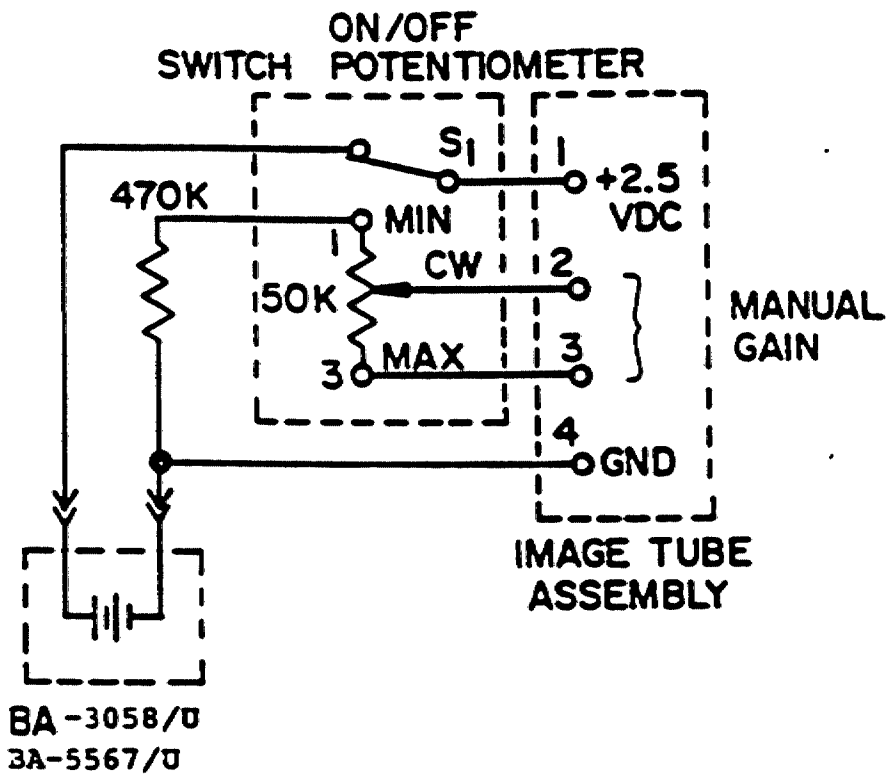
6.6 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.7 Subject term keyword listing.
Night Vision Assembly
Tube Assembly

CUSTODIAN:
Army - CR
Navy - EC

PREPARING ACTIVITY:
Army - CR
Project No. 5855-0023

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NOTES:

1. ON/OFF SWITCH POTENTIOMETER
   A. MANUFACTURER: ALLEN AND BRADLEY, PART NO. GS4T032F503 UA, OR EQUAL
   B. POWER RATING: 0.5 WATT AT 70°C AMBIENT.
   C. RESISTANCE: 50,000 OHMS ± 20 PERCENT.
   D. CURRENT RATING: 1.5 AMPERES FOR 28 VDC.

2. RESISTOR
   A. RESISTANCE: 470,000 OHMS ± 10 PERCENT.
   B. POWER: 1/4 WATT.
   C. TYPE: CARBON, FIXED COMPOSITION

FIGURE 1. Manual brightness gain control wiring diagram.

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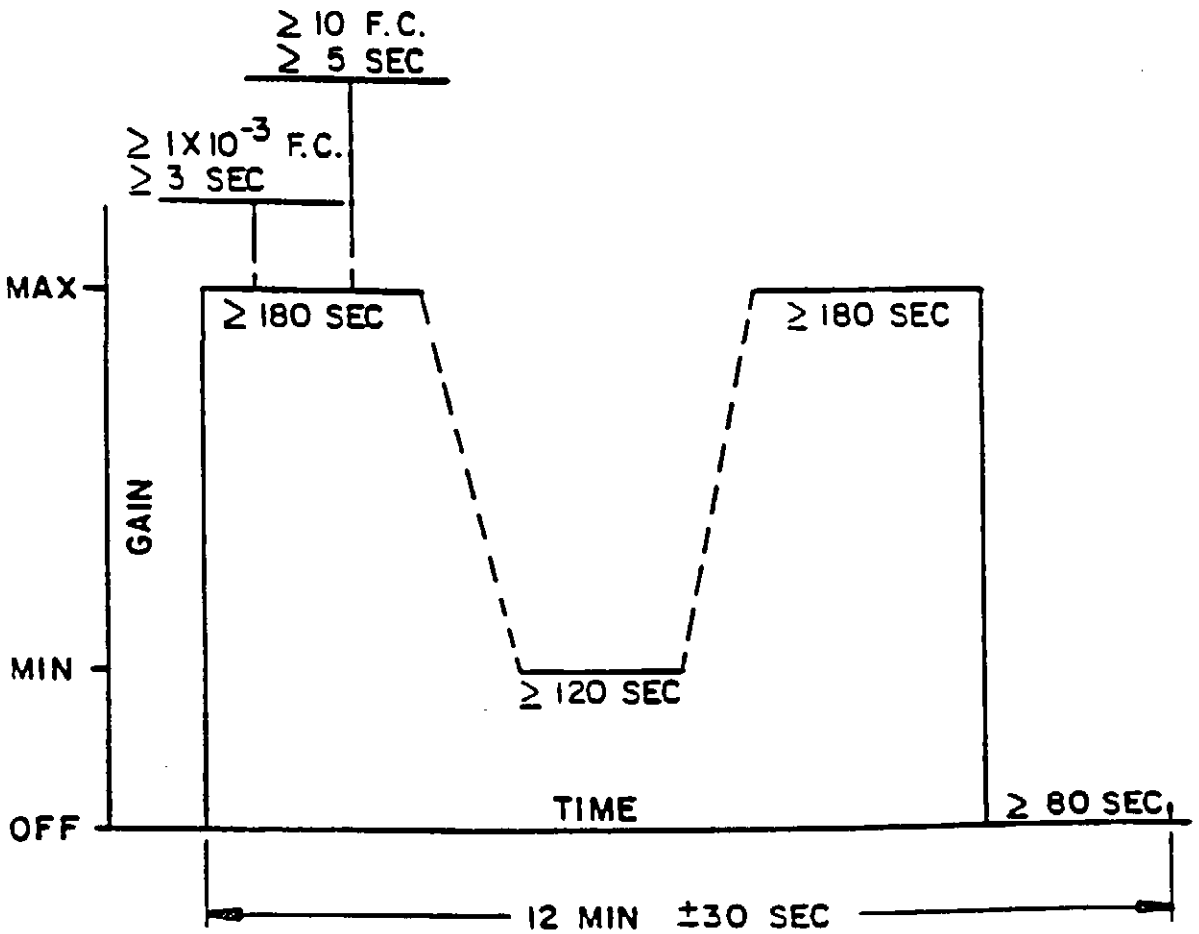


FIGURE 2. Cyclic operation (Burn-In) profile.

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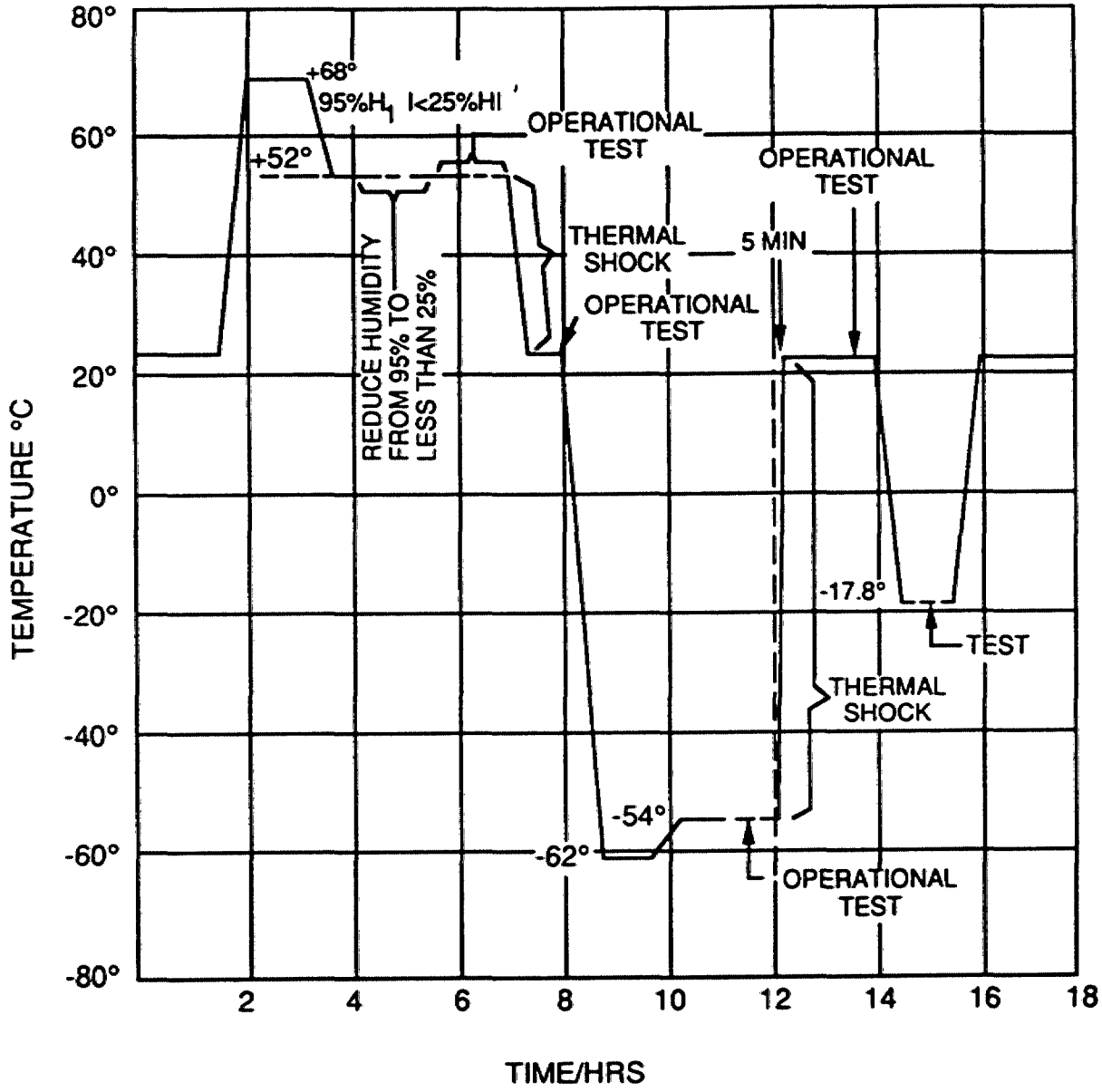


FIGURE 3. Environmental cycle for the assembly.

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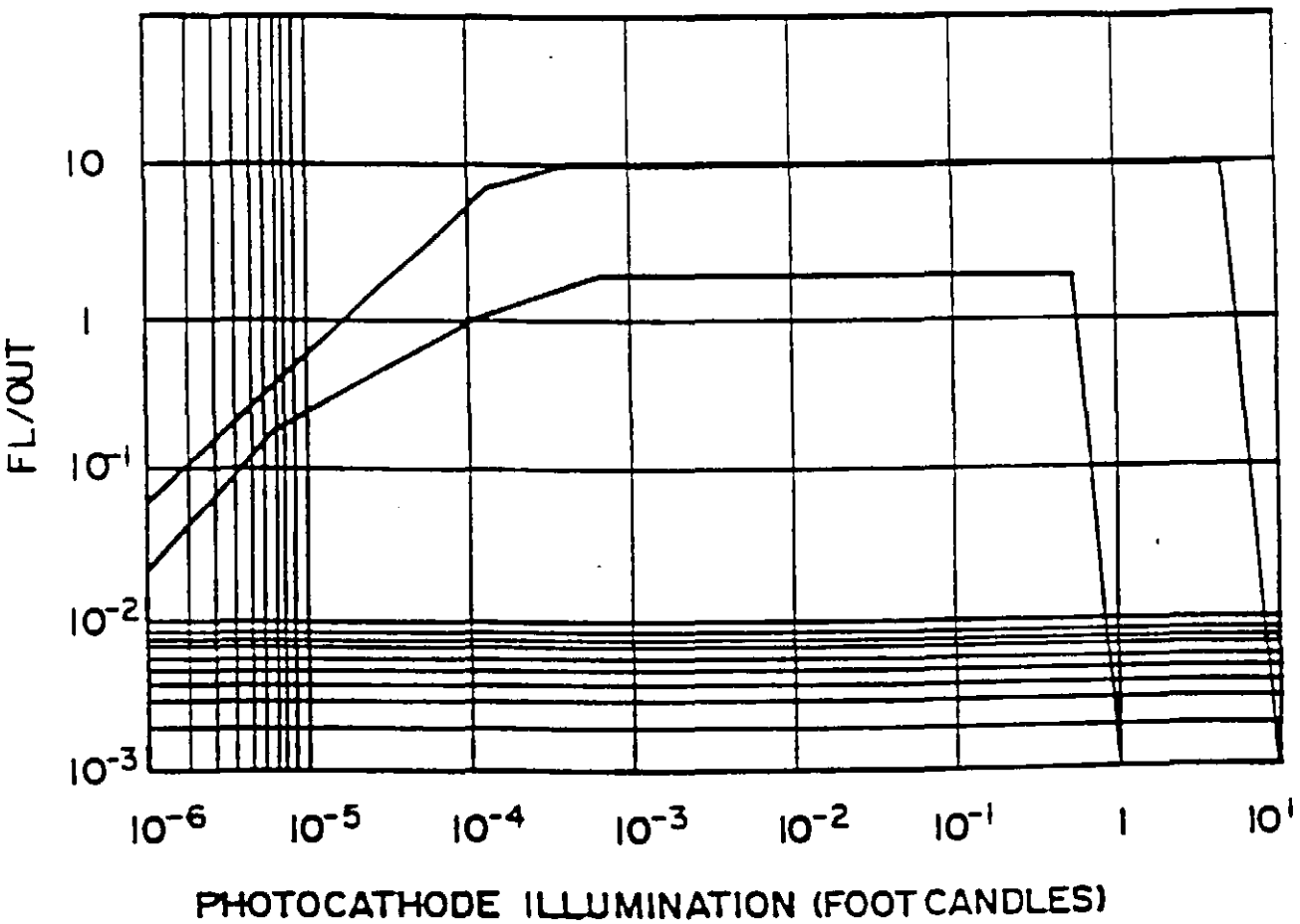


FIGURE 4. Gain/Saturation requirements for tube assembly. Input manual gain setting: Maximum beginning and end of life test.

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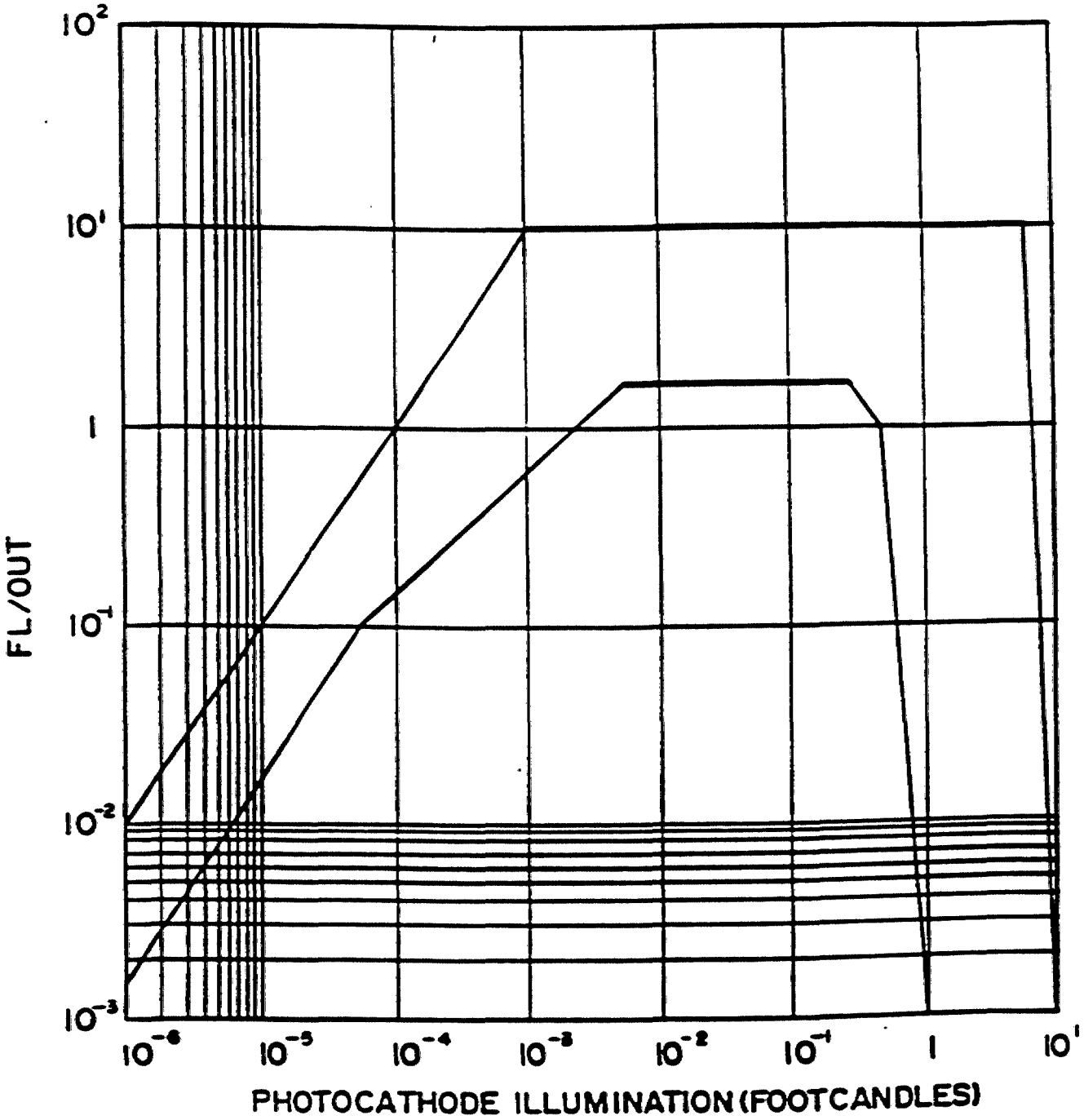


FIGURE 5. Gain/saturation requirements for life test.
Input manual gain settings:
Maximum beginning and end of life test.

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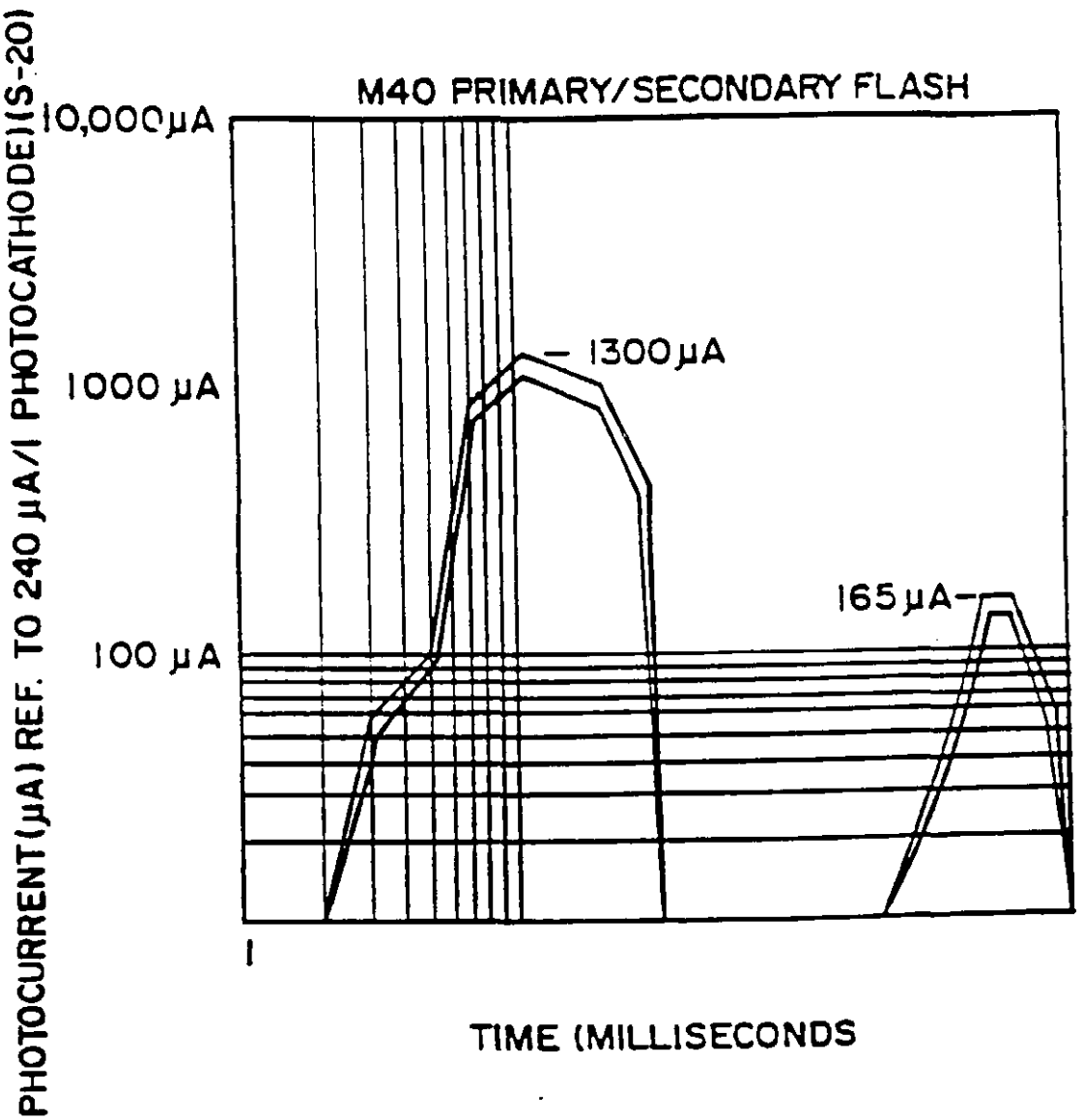


FIGURE 6. Profile (a).

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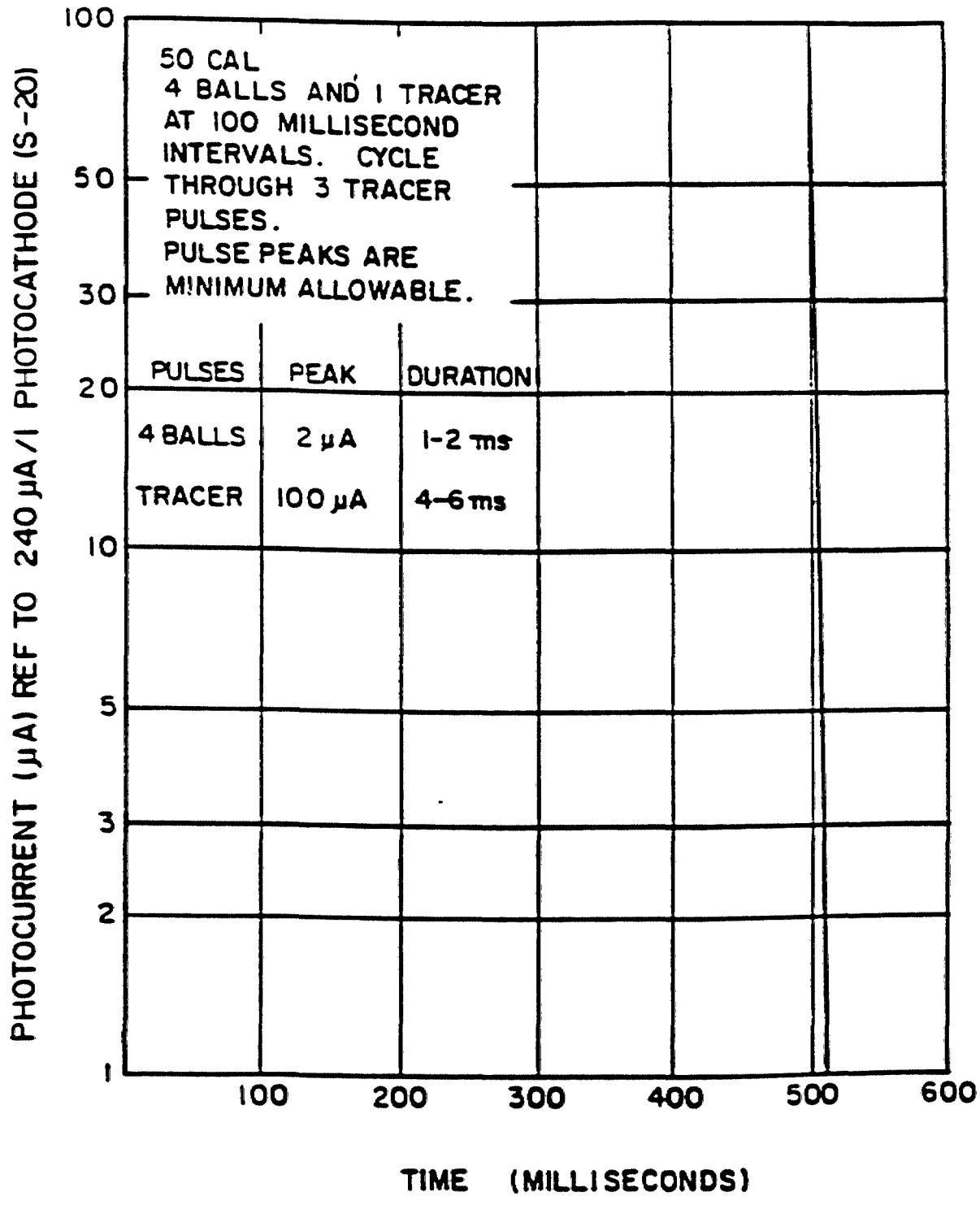


FIGURE 7. Profile (b)

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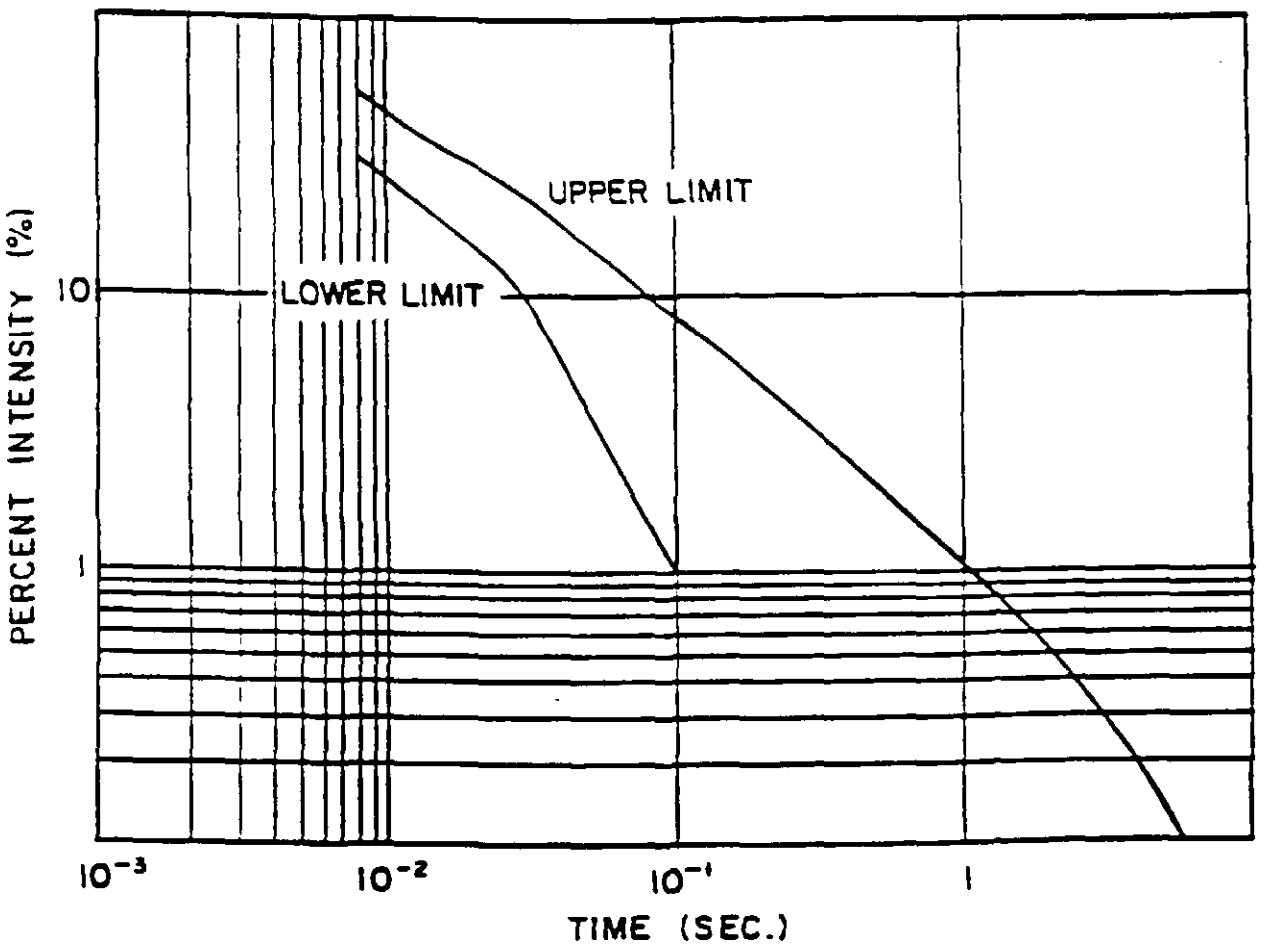
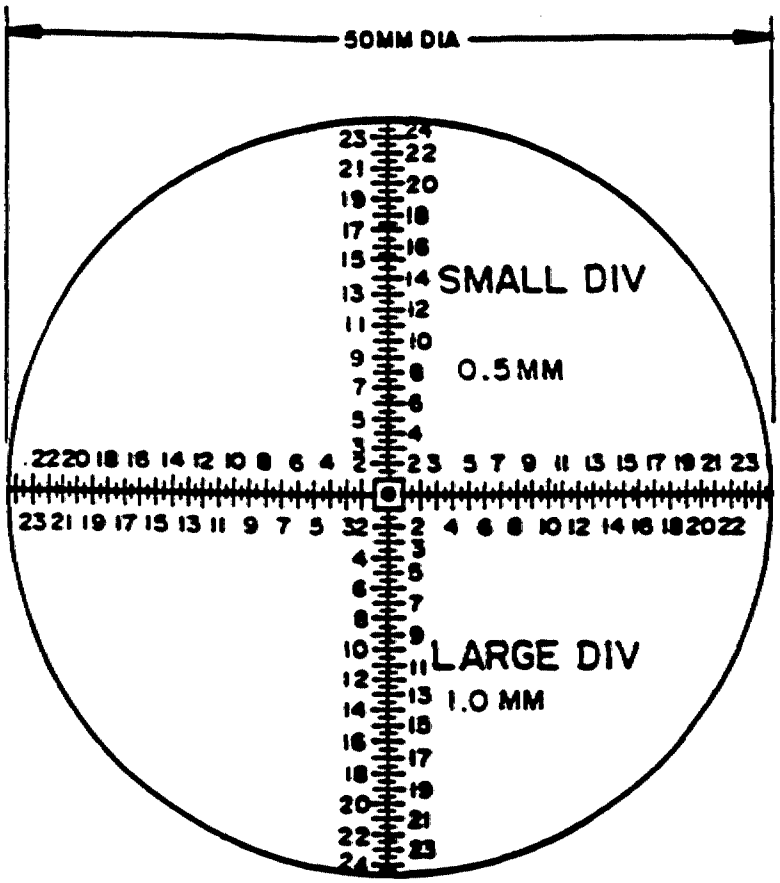


FIGURE 8. Phosphor decay spec. curves

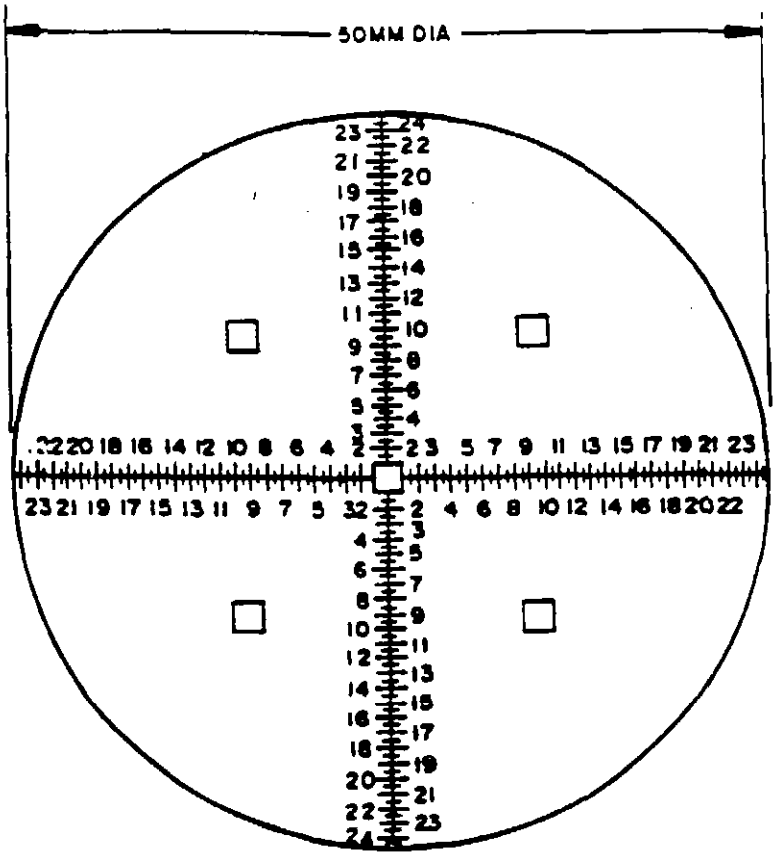
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THE TEST RETICLE SHALL CONSIST OF EQUALLY SPACED LINES FROM THE CENTER OF THE 50 MM DIAMETER CIRCLE TO THE EDGE OF THE CIRCLE IN 4 DIRECTIONS 90° APART. SPACING BETWEEN A LARGE GRADUATION AND A SMALL GRADUATION SHALL BE 0.5 MILLIMETER ±0.03 MILLIMETER. SPACING BETWEEN TWO LARGE GRADUATIONS SHALL BE 1.0 MILLIMETER ±0.03 MILLIMETER. THE CENTER OF THIS TEST RETICLE SHALL BE A SQUARE 2 MILLIMETER ON A SIDE WITH A 0.1270 MILLIMETER DIAMETER SPOT CONCENTRIC WITH THE CENTER OF THE TEST RETICLE. WIDTH OF ALL LINES TO BE 0.127 MILLIMETER ±0.0254 MILLIMETER. ALL LINES, LETTERS AND NUMBERS SHALL BE HIGH CONTRAST BLACK ON A CLEAR GLASS SUBSTRATE 2 INCHES WIDE X 2 INCHES LONG X 0.060 TO 0.10 INCHES THICK. A 25MM target is allowable as long as it meets all of the above requirements.

FIGURE 9. Test reticle

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NOTE:

1. THE TEST RETICLE SHALL CONSIST OF EQUALLY SPACED LINES FROM THE CENTER OF THE 50 MM DIAMETER CIRCLE TO THE EDGE OF THE CIRCLE IN 4 DIRECTIONS 90° APART. SPACING BETWEEN A LARGE GRADUATION AND A SMALL GRADUATION SHALL BE 0.5 MILLIMETER ±0.03 MILLIMETER. SPACING BETWEEN TWO LARGE GRADUATIONS SHALL BE 1.0 MILLIMETER ±0.03 MILLIMETER. THE CENTER OF THIS TEST RETICLE SHALL BE A SQUARE 2 MILLIMETER ON A SIDE WITH A 0.1270 MILLIMETER DIAMETER SPOT CONCENTRIC WITH THE CENTER OF THE TEST RETICLE. WIDTH OF ALL LINES TO BE 0.127 MILLIMETER ±0.0254 MILLIMETER. ALL LINES, LETTERS AND NUMBERS SHALL BE HIGH CONTRAST BLACK ON A CLEAR GLASS SUBSTRATE 2 INCHES WIDE X 2 INCHES LONG X 0.060 TO 0.10 INCHES THICK. A 25MM Target allowable as long as it meets all of the above requirements.
2. THE USAF 1951 RESOLVING POWER TEST TARGET IN THE CENTER SQUARE IS TO BE USED FOR MEASURING CENTER RESOLUTION.
3. THE FOUR EDGE USAF 1951 RESOLVING POWER TEST TARGETS ARE TO BE USED TO MEASURE PERIPHERAL RESOLUTION AND SHALL REFLECT 10 MILLIMETERS FROM CENTER OF PATTERN.

FIGURE 10. 25MM Resolution pattern.

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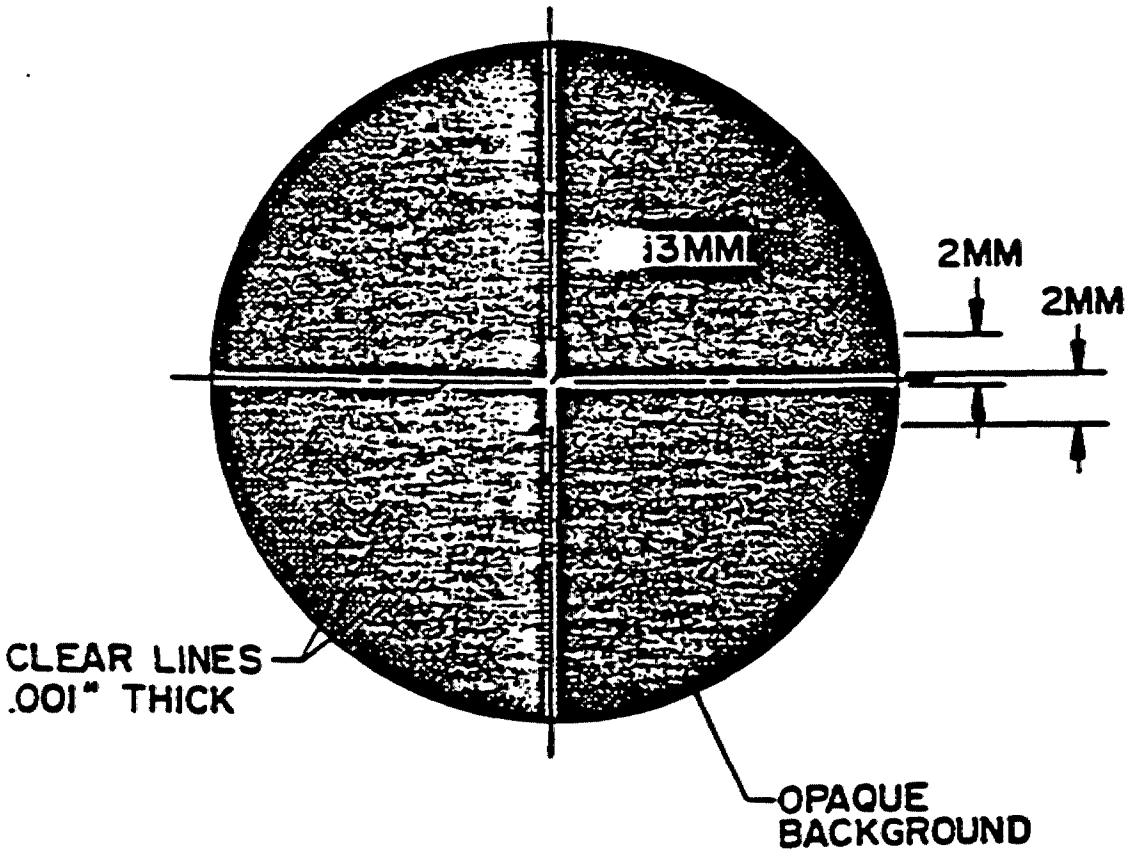


FIGURE 11. Shear distortion test pattern.

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3. DOCUMENT TITLE
IMAGE INTENSIFIER ASSEMBLY 25 MILLIMETER, MICROCHANNEL INVERTER MX-9644/UV

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

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c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

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a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
CECOM	(908) 532-5851	992-5851
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